# CAN SIMPLICITY OVERRIDE PREDICTIVE STRENGTH?

## INTRODUCTION

Neural networks can learn multiple features that are predictive of the same target. However, they do not necessarily use all predictive features equally. Prior work suggests that neural networks can show a **simplicity bias**, preferring simpler predictive features over more complex ones.

This raises a more specific question: **does this preference remain when the complex feature is substantially more predictive?** If a simple feature predicts the target reasonably well, but a more complex feature provides much stronger predictive information, it is unclear whether the network will still prefer the simpler feature.

This work tests whether the preference for the simpler feature persists when the complex feature is made substantially more predictive.

## LITERATURE REVIEW

*The Pitfalls of Simplicity Bias in Neural Networks* (Shah et al., 2020) examines how neural networks choose between predictive features of different complexity. The authors define simplicity as the minimum number of linear pieces a piecewise-linear classifier requires to achieve optimal classification accuracy. Through experiments on synthetic and image-based datasets, they show that neural networks can rely heavily on simpler features while largely ignoring more complex ones.

Although these findings establish that simplicity bias can influence feature selection, an important question remains: how much more predictive must a complex feature become before a neural network begins to favour it over a simpler alternative? Shah et al. demonstrate that simpler features can dominate even when complex features are more predictive, but this does not fully establish how changes in the predictive advantage of a complex feature affect the network's preference. Understanding this relationship could clarify whether simplicity bias remains dominant regardless of predictive strength, or whether there is a point at which the network begins to favour the more complex feature.

This study addresses that question by systematically increasing the predictive strength of a complex feature relative to a simpler one. It investigates whether this change leads to a shift in the feature used by the neural network, and whether a measurable transition point can be identified.

## RESEARCH QUESTION

The principal question of this investigation is whether the neural network still relies on the simpler feature even when the complex feature is more predictive. The threshold is also

explored: how much predictive support the complex feature needs before the model switches to it.

## HYPOTHESIS

The neural network will continue to favour the simpler feature until the complex feature reaches a threshold level of predictiveness. Once that threshold is crossed, the network will begin to favour the complex feature.

## NULL HYPOTHESIS

The neural network does not show a systematic preference for the simpler feature based on its simplicity, and there is no specific threshold of predictiveness at which it switches its preference to the complex feature.

## PROPOSED EXPERIMENT SUITE

Each example has two input features, x_simple and x_complex, and a binary label y. The label is drawn uniformly from {0, 1}. Each feature has its own rule that outputs a binary prediction, and that prediction agrees with y with a chosen probability.

- **Simple feature:** the rule is a threshold at zero, where $x > 0$ means class 1. If the feature should predict 1, x_simple is sampled from U(0, 1). If it should predict 0, it is sampled from U(−1, 0).
- **Complex feature:** the rule is that $\sin(5\pi x) \geq 0$ means class 1. To obtain a value that gives a required prediction, x is sampled from U(−1, 1) and resampled until the rule matches. This makes the rule oscillate several times across the input range, so it is much harder to learn than a single threshold.

**Model and training.** The model is a small MLP with layers $2 \rightarrow 32 \rightarrow 32 \rightarrow 2$ and ReLU activations. It is trained with cross-entropy loss and Adam (learning rate 0.001), with a batch size of 256 for 1,000 epochs. Each setting uses 6,000 training examples and 2,000 test examples. Every value of p_complex is run with three random seeds (42, 43, 44). The seed controls both the data and the weight initialisation.

**Measuring feature reliance.** Test accuracy alone cannot show which feature the network is using, so a separate conflict set of 2,000 examples is built. In each example, the two rules always disagree: the simple rule predicts b and the complex rule predicts $1 - b$. The simple-feature reliance is the fraction of conflict examples in which the model's prediction equals b. A value near 1 means the model follows the simple feature, and a value near 0 means it follows the complex one. Since the two rules always disagree, reliance on the complex feature is 1 − (simple-feature reliance).

**Finding the transition point.** For each p_complex, reliance is averaged over the three seeds and the standard deviation is also recorded. The transition point is where the average reliance falls to 0.5. It is found by locating the first grid value at which reliance drops below 0.5 and linearly interpolating between that value and the previous grid value.

**What is recorded.** For each p_complex, the average test accuracy, the simple-feature reliance and their standard deviations are saved to results.csv. Reliance is also plotted against p_complex with error bars, a dashed line at 0.5 and a dotted line at p_simple, and saved as simplicity_bias_results.png.

**What counts as support.** If reliance stays above 0.5 at p_complex = 0.75 and falls below 0.5 at some larger value, this supports the hypothesis, and the interpolated crossing is the threshold. If reliance is already near or below 0.5 at 0.75, or there is no consistent pattern, this supports the null hypothesis. If reliance never drops below 0.5 even at 1.00, no transition is reported in the tested range.

## CONTROLS AND CONFOUNDING VARIABLES

The simple feature's predictive power is fixed at 75%, while the complex feature's predictive power varies from 75% to 100%. The network architecture, training settings, dataset sizes and evaluation procedure remain constant across experiments. Three random seeds are used to reduce dependence on a single run. A conflict set, in which the two features predict opposite classes, measures reliance on the simple feature independently of ordinary test accuracy. However, the limited number of seeds and possible randomness in training may affect the reliability of the results.

## EXPECTED OUTCOMES AND INTERPRETATION

If the network continues to rely on the simple feature even as the complex feature becomes more predictive, this would support the hypothesis that neural networks may strongly favour simpler features over more predictive ones. If reliance shifts towards the complex feature as its predictive strength increases, this would suggest that predictive strength plays a greater role than simplicity. If the results vary considerably across random seeds or show no consistent pattern, the evidence for simplicity bias would be inconclusive.

## EXPERIMENTAL RESULTS

### 1. Methodology

A synthetic binary classification dataset was generated with 6,000 training samples and 2,000 test samples. The simple feature uses a threshold-based rule, while the complex feature uses a sine-based rule. The simple feature's predictive strength was fixed at 75%, while the complex feature's predictive strength was varied from 75% to 100% in increments of 5%.

A neural network was trained for 1,000 epochs using Adam with a learning rate of 0.001 and a batch size of 256. Each setting was tested with three random seeds. Test accuracy and simple-feature reliance on a conflict set, in which the two features disagree, were recorded.

### 2. Results

The experiment produced the following results:

| Complex-feature predictiveness | Test accuracy | Simple-feature reliance |
|---|---|---|
| 75% | 75.08% | 87.83% |
| 80% | 76.62% | 62.28% |
| 85% | 80.38% | 52.33% |
| 90% | 83.65% | 39.42% |
| 95% | 91.87% | 15.63% |
| 100% | 99.02% | 4.12% |

*Source: Experimental results recorded in the project's results.csv file.*

### 3. Analysis

The results show that the network initially favours the simple feature, relying on it for 87.83% of conflict-set predictions when both features are 75% predictive. However, as the complex feature becomes more predictive, simple-feature reliance steadily decreases. At 90% complex-feature predictiveness, reliance on the simple feature falls below 50%.

These findings suggest that **predictive strength can eventually override the network's initial preference for the simpler feature**.

## LIMITATIONS

The experiment has several limitations. First, using only three random seeds limits the reliability of the results. Second, the simple and complex features differ in their mathematical structure, so differences in learnability may influence feature preference independently of simplicity. Finally, only one network architecture and a fixed set of training settings were tested. Further experiments across different architectures, datasets and training configurations are needed to determine whether these findings hold more broadly.

## CONCLUSION

The results suggest that the model can eventually switch to the complex feature when it is made sufficiently predictive, which was the original research question. However, the fact that the complex feature needed to be so much more predictive before the model switched shows that simplicity bias cannot be ignored.